

## HOW WE TESTED

Since we were testing 18 security suites as exhaustively as it was possible, we decided to use as many PCs as we possibly could to speed up the process. The first step was to prep the PCs for the rigorous testing.

### Prepping the test PCs

We chose four identical PCs for our tests and these were specifically mid-ranged PCs to present a clear use-case scenario. The configuration is as follows:

1. Intel Core 2 Duo E6600
2. Custom HP motherboard
3. 2GB DDR2 RAM @800MHz
4. 80GB SATA HDD

The HDDs were first completely formatted, a copy of Windows 7x64 SP1 was installed on all the machines. Essential apps such as WinRAR, TeraCopy, VLC 2.0, Google Chrome, etc. were installed on all the PCs.

We took a complete system snapshot using Clone-Zilla and each of the four PCs were reset using that very same image to ensure that each and every system was consistent.

We benchmarked the PCs using PCMark vantage (3 runs) to ensure that their performance was within 2 percent of each other.

### Test Process:

The PC was first restored to the system snapshot as specified above and the security suite was installed and allowed to update. All applicable optional components were installed, including toolbars and Windows gadgets provided by the suite.

The test was divided into two main parts, namely, performance and productivity.

The productivity tests were run first, followed by the performance tests. Detailed notes were taken during the entire test process.

After each security suite was tested, the system was again restored to the system snapshot

### Productivity tests

These tests were designed to measure how much a system is affected by the presence of the security suite.

1. **Benchmark:** We ran the peace-keeper benchmark (<http://dgit.in/AVSuite>) on the test system before and during a system scan. The benchmark measures various aspects of a browser's performance, including video playback and image-editing. We found that this benchmark scales well across a variety of loads and gave consistent results. Maximum weightage was given to this test as it tries to mimic a real world usage scenario.
2. **File transfer:** We transferred 2 GB of assorted files from partition to partition and compared it to the time taken to transfer the same files before the security suite was installed.
3. **File Compression:** We compressed the above mentioned assorted files into a .zip file using 7-zip and its default options.
4. **Total system scan:** A full system scan was run on the test PC, all scan modes except a memory scan (RAM) were enabled, where applicable. This was mainly because not all suites had the option to scan RAM and we wanted consistent results.
5. **Boot time:** This test was a measure of the AV suite's impact on the boot time of our test systems. The results of this test were given a minimum weightage as this score is largely irrelevant unless you spend a lot of time shutting down and restarting your PC.

Boot time was measured using the "Windows Performance Analysis Tool", which is part of the Windows SDK.

### Performance tests

In these tests, our aim was to measure the performance of the AV suite in terms of its detection rate and ability to detect and prevent malware and rootkits from accessing the system. All these tests were performed with all settings at their default values to measure out-of-the-box performance of every suite.

1. **Virus samples:** We downloaded a selection of 9974 viruses, trojans, etc. from an online virus vault. These samples covered a period from the 1st to the 15th of March and were put on a formatted pen-drive and the AV suite was allowed to run a full scan on the drive. At the end of the test, the log file was used to verify the number of infected files actually detected by the security suite.
2. **Malware domains:** To test the effectiveness of the link-scanner which came bundled with almost all the suites that we had for testing, we compiled a list of infected domains, featuring a variety of malware including fake AV software. The response of the AV suite to these sites were noted and rated
3. **Payloads:** A payload is that part of a computer virus which performs some malicious action. We created 13 (this wasn't intentional) payloads using the Metasploit Framework. The payloads were hosted on a server running *backtrack* and metasploit and the test PC had to navigate

to the server's IP and then download the files. The payloads were divided into 3 tiers based on the challenge they would present to the AV engine. The suites were rated on their response to these payloads.

- a. Tier 1: These payloads utilize nearly all known virus bypassing techniques known to humankind. We inserted junk code in the program, encoded multiple times using various encryption schemes. This was followed by a lengthening of the decryption loop (to understand why, read Digit 101). We then compiled the code using the 32-bit *mingw compiler*, and stripped the debugging symbols to further throw off the engines. These payloads were the most difficult to detect and usually need excellent heuristics scanners to detect.
- b. Tier 2: We used an interesting method to get around the heuristics engine here. We can't go into much detail, but we basically XORed the shellcode and the decryption loop required input from the command line, which throws off AV engines. Otherwise it employed the usual encryption schemes. These samples lie somewhere in between hard and easy to detect.
- c. Tier 3: The simplest of payloads to be detected, they employed the least amount of encoding. But the change here was that these samples utilized backdooring and they were embedded into harmless looking executables - VLC 2.0 setup, Microsoft Malicious Software Tool setup and 7zip 32-bit setup. Still most of the security suites detected samples from this tier.